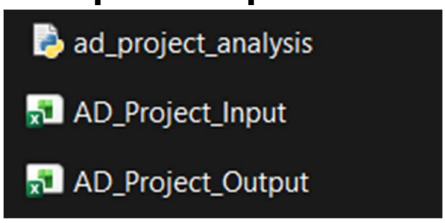
Complete Explanation of Input and Output Excel Files



Your project uses:
1. Input Excel File - "AD_Project_Input.xlsx"
and
2. Output Excel File – "AD_Project_Output.xlsx"
The input file gives raw experimental and process data to Python.
The output file gives final calculated results used for:

- thesis discussion
- Aspen Plus input
- model validation
- optimization work

PART 1 — INPUT EXCEL FILE
File Name:
AD_Project_Input.xlsx
This contains multiple sheets.

SHEET 1 — Daily_Methane

Purpose
This sheet stores:
daily cumulative methane production
from your BMP experiment.
This is the most important experimental data.

Example Table

Day	CH4_No_Biochar	CH4_5pct_Biochar	CH4_2pct	CH4_8pct	CH4_10pct
1	12	18	14	16	13
2	25	36	28	31	27
3	42	58	47	52	44
…	…	…	…	…	…
10	320	429	362	397	366

Meaning of Each Column
Day
Digestion day
Example:
Day 1 to Day 30

CH4_No_Biochar
Methane production without biochar
This is:
control experiment
Used as baseline.

CH4_5pct_Biochar
Methane production with fixed biochar dose
Example:

5%
Usually the best-performing case.

CH4_2pct, 8pct, 10pct
Methane production at different biochar dosages
Used for:
optimum dose selection

Why This Sheet Is Needed
Used for:

- methane yield calculation
- optimum biochar dose
- alpha calculation
- Gompertz fitting
- Aspen validation

Most important sheet.

SHEET 2 — Ultimate_Analysis

Purpose
This sheet stores:
elemental composition
of all non-conventional (NC) components.
Used for:

- Buswell equation
- Aspen NC component definition

Example Table

Material	C_wt%	H_wt%	O_wt%	N_wt%	Ash_wt%
FoodWaste	48.0	6.4	37.5	2.1	6.0
SB_Biochar_500	72.0	2.5	12.0	0.8	12.7
Digest	32.0	4.1	28.0	2.5	33.4

Meaning of Each Column
Material
Name of NC component:

- food waste
- sugarcane bagasse biochar
- digest

C, H, O, N
Ultimate analysis values
These are used in:
Buswell equation
for methane prediction.

Ash
Non-biodegradable solid content
Used for:
Aspen NC definition

Why This Sheet Is Needed
Used for:

- theoretical methane potential
- substrate validation

- RYIELD logic
- Aspen component setup

Very important.

SHEET 3 — Process_Inputs

Purpose
Stores:
model constants and assumptions
Used in Python and Aspen.

Example Table

Parameter	Value	Unit
VS_added	1.0	gVS
Optimum_Biochar_Dose	0.05	fraction
Base_k	0.25	day^{-1}

Meaning of Each Row

VS_added
Volatile solids added during BMP
Used in:
methane yield calculation

Optimum_Biochar_Dose
Best biochar amount
Example:
5% = 0.05
Used in:
alpha calculation
and
Aspen simulation

Base_k
Initial kinetic constant
Used for:
RCSTR reactor
before biochar enhancement.

Why This Sheet Is Needed
Used for:

- alpha calculation
- effective k calculation
- RCSTR setup

Very important.

PART 2 — OUTPUT EXCEL FILE
File Name:
AD_Project_Output.xlsx
This is generated automatically by Python.
This contains final calculated results.

SHEET 1 — Methane_Yield

Purpose
Shows methane yield comparison for all cases.

Example Table

Case	Methane_Yield
No Biochar	320
2% Biochar	362
5% Biochar	429
8% Biochar	397
10% Biochar	366

Why Important
This proves:
biochar improves methane production
and identifies:
optimum biochar dose

SHEET 2 — Alpha_Calc

Purpose
Shows:
biochar enhancement factor

Example Table

Parameter	Value
Y_without	320
Y_with	429
Biochar_Dose	0.05
Alpha	6.81

Why Important
Because:
biochar is unreacted
its effect must be represented mathematically.
This α is the heart of your thesis.

SHEET 3 — Kinetic_k

Purpose
Shows:
RCSTR effective kinetic constant

Example Table

Parameter	Value
Base_k	0.25
Effective_k	0.335

Why Important
This is the actual value used in:
Aspen RCSTR Power Law kinetics
This connects Python to Aspen.
Very important.

SHEET 4 — RYIELD_Table

Purpose
Defines decomposition in Aspen RYIELD.

Example Table

Product	Fraction
Acetate	0.44
H2	0.07
CO2	0.19
H2O	0.15
Digest	0.15

Why Important
This tells Aspen:
FOODWASTE
→ intermediates
and NOT directly methane.
This makes model scientifically correct.

Python Algorithm with Code Explanation

Biochar Effect on Methane Production in AD
This Python script is used to convert your lab experimental data into:

- methane yield comparison
- optimum biochar dose
- biochar enhancement factor (α)
- effective kinetic constant (k_{eff})
- RYIELD distribution for Aspen

This creates the connection between:
Lab Work → Python → Aspen Plus

Full Algorithm Flow
Start
↓
Read Input Excel File
↓
Extract Final Methane Data
↓
Compare Methane Yield for All Biochar Doses
↓
Find Best Biochar Dose
↓
Calculate Alpha (α)
↓
Calculate Effective k
↓
Prepare RYIELD Distribution Table
↓
Export Final Results to Output Excel
↓
End
This is the complete algorithm.

STEP 1 — Import Required Libraries

Code

```
import pandas as pd
```

Why
We use:
pandas
because it helps to:

- read Excel files
- process data
- create result tables
- export Excel output

It is the most important library for this work.

STEP 2 — Define File Names

Code

```
input_file = "AD_Project_Input.xlsx"
output_file = "AD_Project_Output.xlsx"
```

Why

Python must know:

- where to read input
- where to save output

Simple but necessary.

STEP 3 — Read Input Excel Sheets

Code

```
methane = pd.read_excel(input_file, sheet_name="Daily_Methane")
process = pd.read_excel(input_file, sheet_name="Process_Inputs")
```

What It Does

Reads:

Daily_Methane

and

Process_Inputs

from Excel.

Why

Because all calculations depend on these sheets.
Without input data, nothing can be calculated.

STEP 4 — Extract Final Methane Values

Code

```
final_row = methane.iloc[-1]
```

What It Does

This selects:

last row

from Daily_Methane sheet.

Example:

Day	No Biochar	5% Biochar
10	320	429

This is:

final cumulative methane production

Why

Final methane value is needed for:

- methane yield comparison
- alpha calculation
- validation

Very important step.

STEP 5 — Create Methane Yield Table

Code

```
yield_table = pd.DataFrame({
    "Case": [
        "No Biochar",
        "2% Biochar",
        "5% Biochar",
        "8% Biochar",
        "10% Biochar"
    ],
    "Methane_Yield_mL_CH4_gVS": [
        final_row["CH4_No_Biochar"],
        final_row["CH4_2pct"],
        final_row["CH4_5pct_Biochar"],
        final_row["CH4_8pct"],
        final_row["CH4_10pct"]
    ]
})
```

What It Does

Creates:

Case	Methane Yield
No Biochar	320
5% Biochar	429

etc.

Why

To compare:

which biochar dose gives maximum methane

This helps find:

optimum biochar dose

STEP 6 — Read Required Parameters

Code

```
y_without = final_row["CH4_No_Biochar"]
y_with = final_row["CH4_5pct_Biochar"]

B = float(
    process.loc[
        process["Parameter"] == "Optimum_Biochar_Dose",
        "Value"
    ].values[0]
)
```

What It Does

Reads:

- methane without biochar
- methane with biochar
- biochar dose

Example:

Ywithout = 320

Ywith = 429

B = 0.05

Why
These values are needed for:
alpha calculation

STEP 7 — Calculate Alpha (α)

Formula

$$\alpha = \frac{\left(\frac{Y_{with}}{Y_{without}} - 1\right)}{B}$$

Code

```
alpha = ((y_with / y_without) - 1) / B
```

Using Your Data

$$\alpha = \frac{\left(\frac{429}{320} - 1\right)}{0.05}$$

Result:
α = 6.81

Why
Because:
biochar remains unreacted
so its effect must be represented mathematically.
This is the strongest scientific point.

STEP 8 — Save Alpha Table

Code

```
alpha_table = pd.DataFrame({
    "Parameter": [
        "Y_without",
        "Y_with",
        "Biochar_Dose",
        "Alpha"
    ],
    "Value": [
        y_without,
        y_with,
        B,
        alpha
    ]
})
```

Why
This makes results clear and ready for thesis.

STEP 9 — Calculate Effective k

Formula

$$k_{eff} = k(1 + \alpha B)$$

Code

```
base_k = float(
    process.loc[
        process["Parameter"] == "Base_k",
        "Value"
    ].values[0]
)

k_eff = base_k * (1 + alpha * B)
```

Using Your Data

$$k = 0.25$$

$$k_{eff} = 0.25(1 + 6.81 \times 0.05)$$

Result:
k_eff = 0.335

Why
This is the value used in:
Aspen RCSTR
for biochar case.
This directly connects Python to Aspen.

STEP 10 — Save Kinetic Table

Code

```
k_table = pd.DataFrame({
    "Parameter": [
        "Base_k",
        "Effective_k"
    ],
    "Value": [
        base_k,
        k_eff
    ]
})
```

Why
This gives final RCSTR input values.

STEP 11 — Create RYIELD Table

Code

```
ryield = pd.DataFrame({
    "Product": [
        "Acetate",
        "H2",
        "CO2",
        "H2O",
        "Digest"
    ],
    "Fraction": [
        0.44,
        0.07,
        0.19,
        0.15,
        0.15
    ]
})
```

Why

Aspen RYIELD needs:
FOODWASTE
→ intermediates
and not direct methane.
This makes model scientifically correct.

STEP 12 — Export Final Output Excel

Code

```
with pd.ExcelWriter(output_file) as writer:
```

and save all sheets.

Why

Final output file becomes:
ready for thesis
ready for Aspen input
Very important final step.

Final Output Sheets

Methane_Yield
Alpha_Calc
Kinetic_k
RYIELD_Table

Aspen Plus Simulation

Python provides:

- optimum biochar dose
- methane yield data
- biochar enhancement factor α
- effective kinetic constant k_{eff}
- RYIELD decomposition fractions

Aspen uses these values to simulate:
complete anaerobic digestion process
and validate the methane production.
This creates the final connection:
Lab Work → Python → Aspen Plus → Final Validation

Why We Use Aspen Plus
Lab experiment gives:
real methane production
but it cannot easily show:

- internal reaction pathways
- gas-liquid distribution
- reactor behavior
- effect of kinetics on methane formation
- optimization of operating conditions

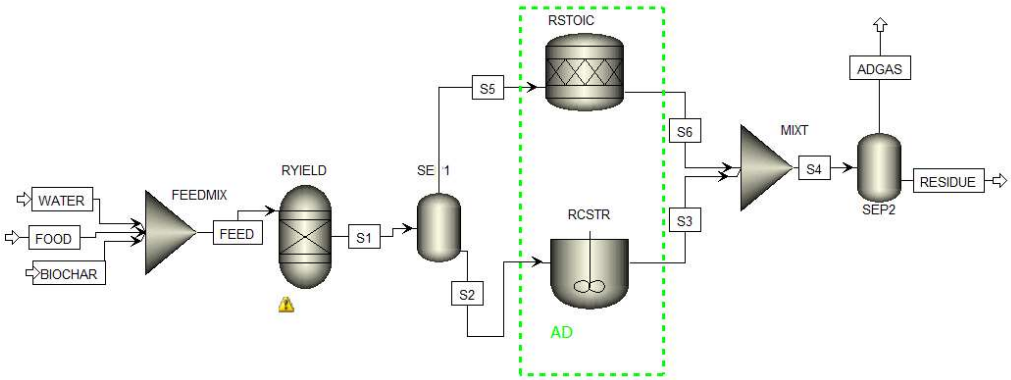
Aspen helps to simulate all these scientifically.
It makes the thesis strong and publishable.

Important Scientific Assumption
Biochar does NOT react
Biochar is treated as:
non-conventional (NC) unreacted additive
Its role is only:
kinetic enhancement
This means:

- no reaction in RYIELD
- no reaction in RSTOIC
- no reaction in RCSTR stoichiometry

Only:
RCSTR kinetic constant changes
This is the most important part of your model.

Final Aspen Flowsheet



This is the final correct model.

Step 1 — Component Definition

Conventional Components
These are normal components:

- H_2O
- CH_4
- CO_2
- H_2
- Acetic Acid (or ACETIC)
- NH_3 (optional)

These are used in reactors.

Non-Conventional Components
These are:

- FOODWASTE
- BIOCHAR
- DIGEST

These are defined using:

- proximate analysis
- ultimate analysis

from your lab data.

Why
Because food waste and biochar are not pure chemicals.
They are solid complex materials.
So Aspen uses:
NC components
for realistic simulation.

Step 2 — Property Method
Use:
NRTL
or
ELECNRTL (optional)
Recommended:
NRTL
for thesis simplicity.

Why
Because AD contains:

- water
- organic acids
- gases
- non-ideal liquid behavior

NRTL handles this well.

Step 3 — FEEDMIX Block

Purpose
Mix:

- food waste
- water
- biochar

to create:
FEED stream

Why
This represents the digester feed.
Water provides slurry condition.
Biochar enters here and stays unreacted.

Step 4 — RYIELD Block

Purpose
Represents:
Hydrolysis + Acidogenesis
Food waste is decomposed into intermediates.
Reaction:
FOODWASTE
→ Acetate + H_2 + CO_2 + H_2O + Digest
Important:
NO methane here

Why
Real AD happens in stages:

1. Hydrolysis
2. Acidogenesis
3. Acetogenesis
4. Methanogenesis

Methane should form later.
This makes model scientifically correct.

RYIELD Input Fractions
Example:

Product	Fraction
Acetate	0.44
H_2	0.07
CO_2	0.19
H_2O	0.15
Digest	0.15

These come from:

- Buswell equation
- COD balance
- BMP data
- literature validation

Step 5 — SEP1 Block

Purpose
Separate RYIELD output into:
Gas Stream → S5
Contains:

- H_2
- CO_2

Liquid Stream → S2
Contains:

- Acetate
- Water

- Digest
- Biochar

Why
Because methane forms by:
two different pathways
So both must be treated separately.
Very important.

Step 6 — RSTOIC Block

Purpose
Represents:
Hydrogenotrophic Methanogenesis
Reaction:

$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

Why
Gas-phase H2 and CO2 must be converted to methane.
This is one major methane pathway.
Simple and strong.

Setup
Use:
Fractional conversion
Example:
70–90%
depending on experimental support.

Step 7 — RCSTR Block

Purpose
Represents:
Acetoclastic Methanogenesis
Reaction:

$$CH_3COOH \rightarrow CH_4 + CO_2$$

This is the main methane production step.
Most methane comes from here.

Why
Acetate is the major methane precursor in AD.
This is where:
biochar effect is modeled
Most important block.

RCSTR Kinetic Equation
Conceptually:

$$r = k(1 + \alpha B)C$$

where:

- k= base kinetic constant
- α= biochar enhancement factor
- B= biochar dose
- C= acetate concentration

Aspen uses simplified form:

$$r = k_{eff}C$$

where:

$$k_{eff} = k(1 + \alpha B)$$

Example
From Python:
Base k = 0.25
Alpha = 6.8
B = 0.05
Then:
k_eff = 0.335
This is entered in:
Power Law kinetics
inside RCSTR.

Your Current Setup
From your screenshot:
k = 0.35
n = 0
E = 0
Recommended improvement:
k = 0.35
n = 1
E = 0
because:

$$r = kC^n$$

and first-order reaction is more realistic.

Step 8 — MIXT Block

Purpose
Mix methane from:

- RSTOIC
- RCSTR

to create:
total biogas stream

Why
Final methane comes from both pathways.
So both must be combined.

Step 9 — SEP2 Block

Purpose
Separate final stream into:
ADGAS
Final methane-rich biogas
RESIDUE
Digestate + water + biochar

Why
To obtain final usable product stream.
This matches real digester output.

Final Outputs

ADGAS
Contains:

- CH_4
- CO_2
- trace gases

Used for:
methane yield validation

RESIDUE
Contains:

- digest
- water
- unreacted biochar

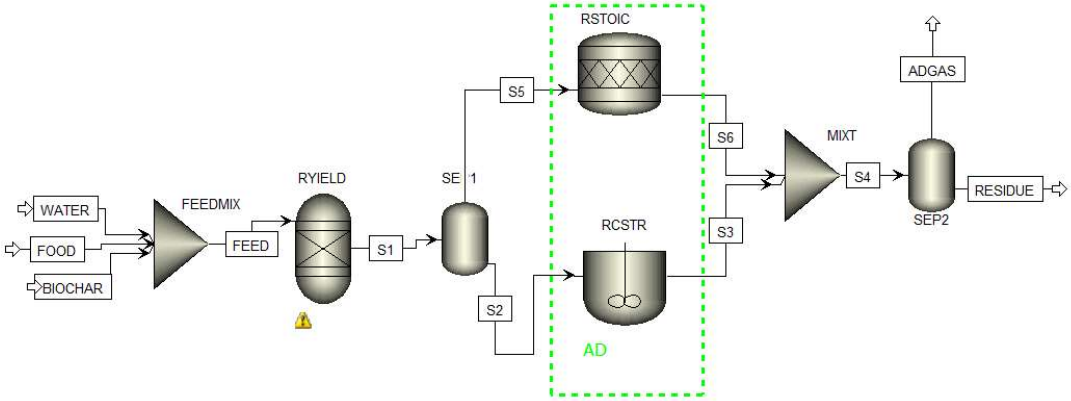
This proves:
biochar remains unreacted
Very strong thesis point.

Model Validation
Compare:
Aspen methane yield
with
experimental methane yield
Example:

Source	Methane Yield
Lab BMP	429
Aspen	417

Good agreement means:
model is valid

Final Scientific Conclusion
Biochar does not participate in chemical reactions.
It improves methane production only by increasing:
methanogenesis reaction kinetics
inside RCSTR.
This is the strongest scientific contribution of your work.


RSTOIC
ADGAS
S5
S6
MIXT
RYIELD
SE 1
S4
RESIDUE
WATER
FEEDMIX
FOOD
FEED
S1
RCSTR
S3
SEP2
BIOCHAR
S2
AD